Firewood Lift

A simple bolt-together goods lift for the back porch. Roll the loaded hand truck on, lift it 30 inches, roll it off. 120 V electric hoist. Version 1.0, October 2026.

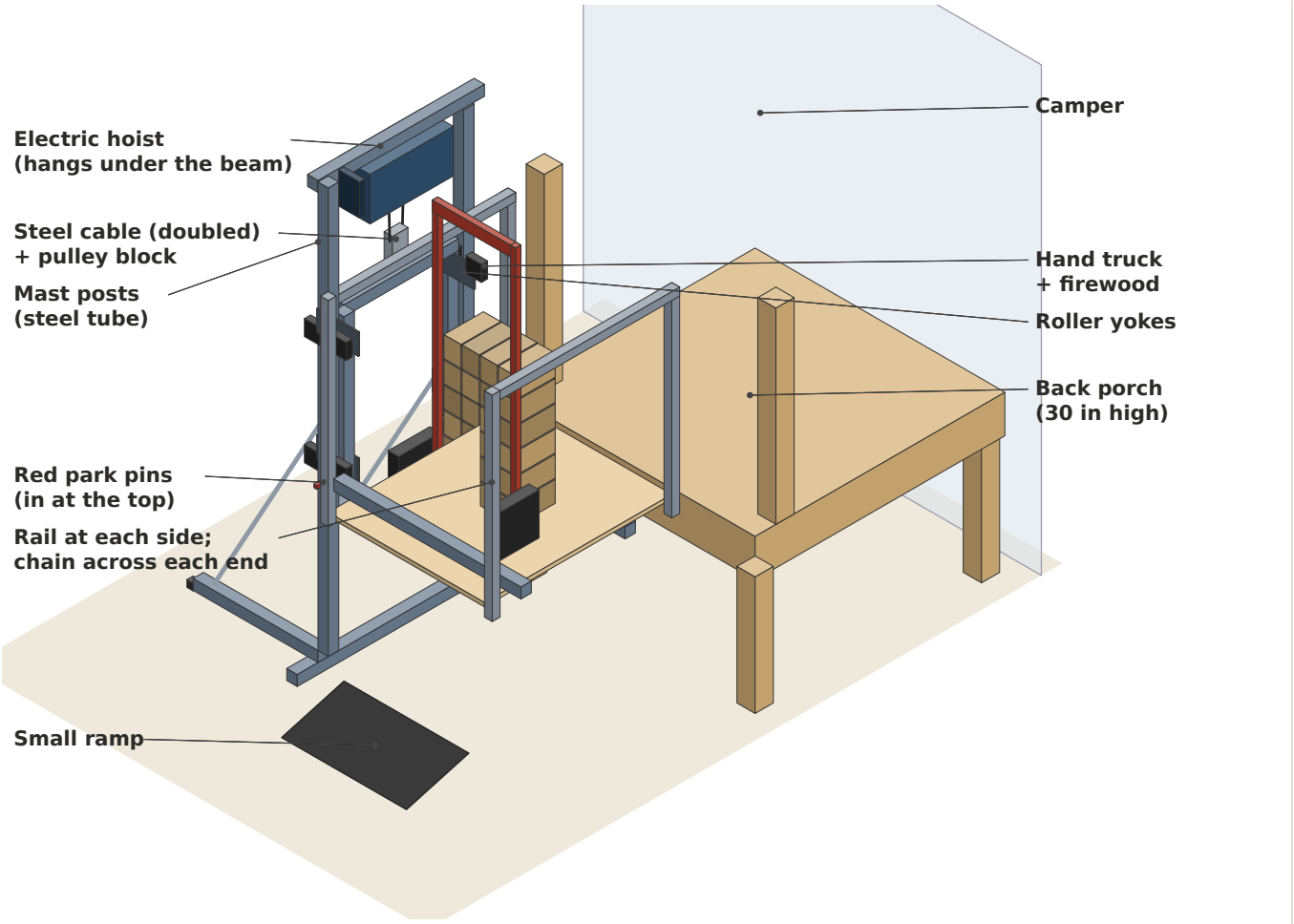


Figure 1: The lift at porch level with a loaded hand truck. The red park pins are in, so the platform rests on them, not the cable, while you roll the hand truck off.

FOR FIREWOOD ONLY. NO RIDERS, EVER. This lift hangs on a cable. It is not for people: not Nadine, not kids, not you. Put a "NO RIDERS" sign on it. Never put a hand or foot under the platform. 300 lb max on the platform, including the hand truck.

1. How it works

- An **electric hoist** hangs under the top beam. That's a winch made for lifting: it has a brake that holds the load when you let go, and a limit switch at the top.
- The cable runs down to a pulley on the platform and back up. Doubling the cable halves the pull, so the hoist only works at about half its rating.
- Rollers on two steel posts keep the platform level, so the hand truck can't tip.
- At the top you put in two **park pins**. The platform sits on the pins while you roll the hand truck off, so it can't drop even if the cable failed.
- About 10 seconds up.

> **Why a hoist and not an ATV or truck winch:** those are pulling winches. Their brakes aren't made to hold a hanging load and their makers say not to lift with them. An electric hoist costs about the same and is built for this job.

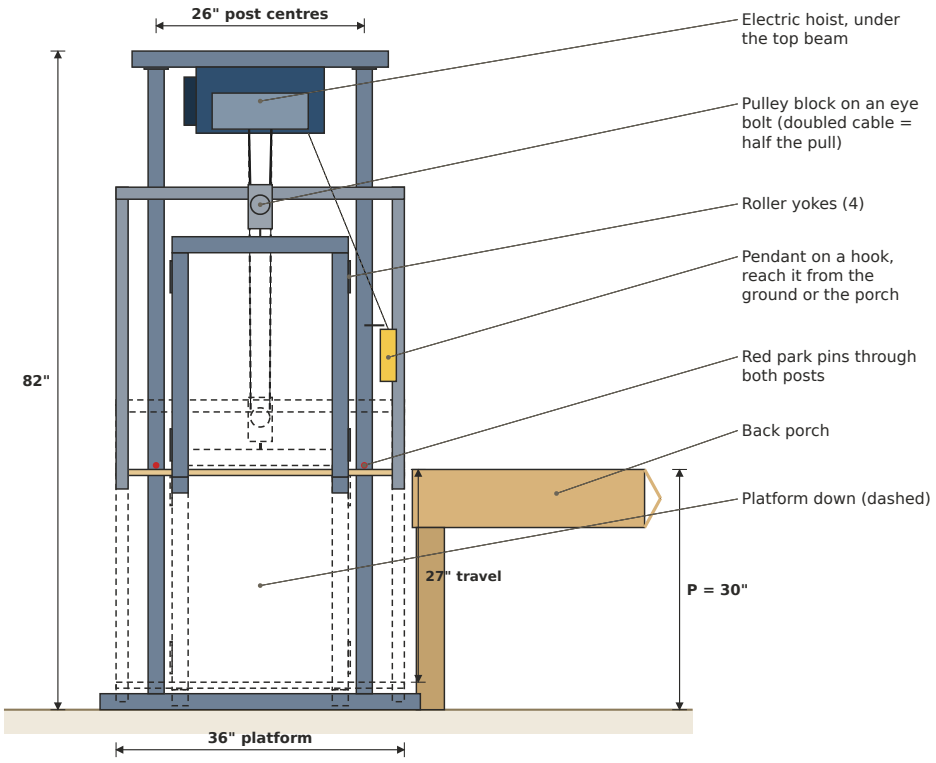


Figure 2: Seen from behind the mast. The platform is drawn at porch level (solid) and on the ground (dashed).

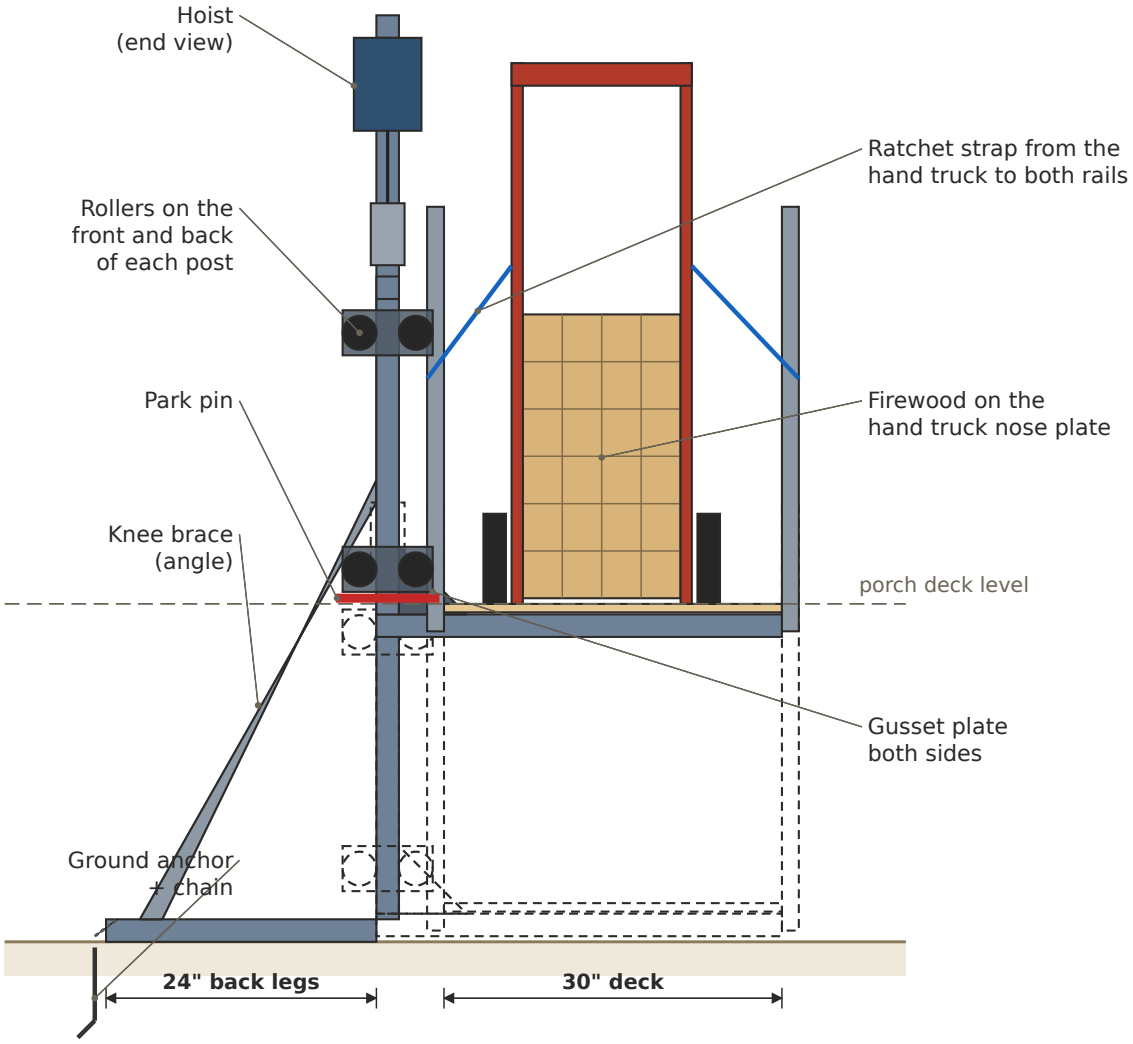
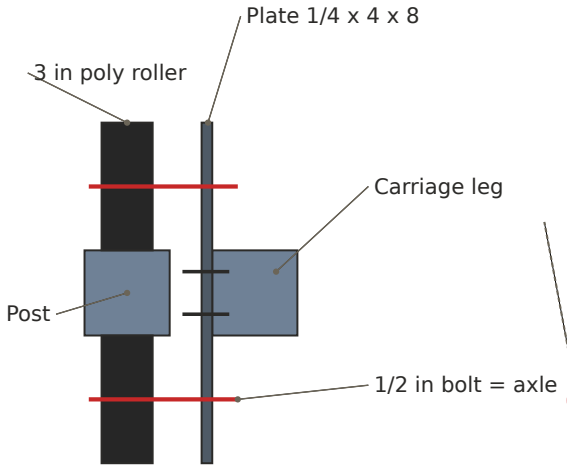


Figure 3: Side view from the porch end, platform at the top with a loaded hand truck. The ground anchors stop the mast tipping forward.

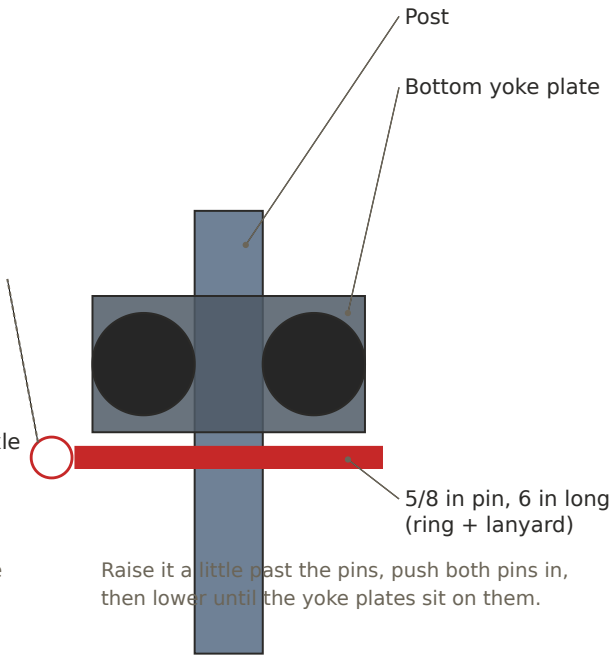
2. Detail drawings

A. Roller yoke, seen from above



Each roller just touches the post. Front rollers take the push at the bottom, back rollers the pull at the top.

B. Park pin (at the top)



Raise it a little past the pins, push both pins in, then lower until the yoke plates sit on them.

C. Pulley block on the carriage

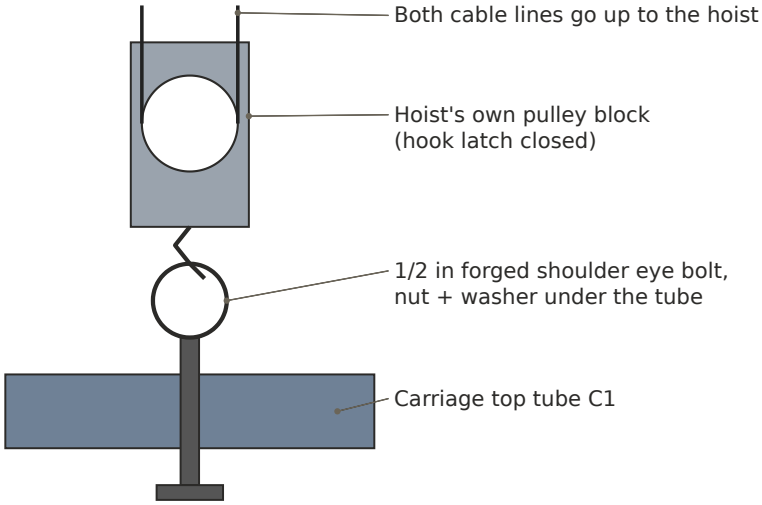
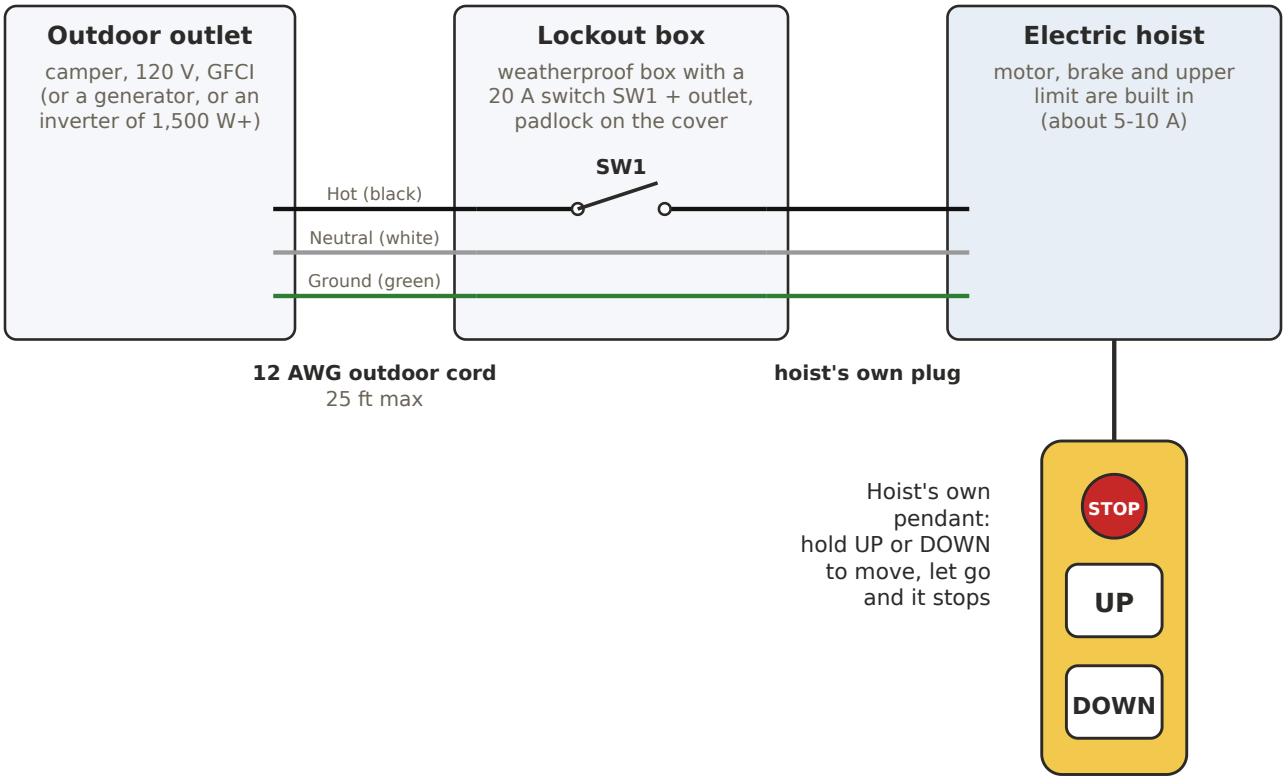


Figure 4: The roller yokes, the park pins and the pulley block.

3. Wiring



How to use it:
1. Turn SW1 on (unlock the cover).
2. Hold UP or DOWN on the pendant. Let go and the brake holds it.
3. When you're done, lower it to the ground, turn SW1 off and lock the cover, so nobody can run it while you're away.

Only the black (hot) wire goes through SW1. The green ground wire must connect all the way from the outlet to the hoist. Don't open up the hoist or change its pendant: it already has its own stop button and limit.

Figure 5: Wiring. The hoist plugs in like a power tool. The only thing you add is a lockout switch.

4. Cut lists

2" x 2" square steel tube, 1/8" wall

ID	Piece	Qty	Length	Notes
B1	Base rail	1	40"	Lies on the ground under the mast
B2	Back legs	2	24"	Stick out behind the mast; anchors at the ends
P1	Posts	2	78"	Rollers run on the front and back faces: keep them smooth
T1	Top beam	1	32"	The hoist hangs under it
A1	Platform arms	2	36"	Carry the deck
L1	Carriage legs	2	28"	Stand on the back ends of the arms
C1	Carriage top tube	1	22"	Across the leg tops; eye bolt in the middle

Buy two 20 ft sticks. Stick 1: P1, P1, B1, A1. Stick 2: A1, T1, L1, L1, B2, B2, C1.

1-1/2" x 1-1/2" x 3/16" steel angle (two 20 ft sticks)

ID	Piece	Qty	Length	Notes
K1	Knee braces	2	44"	Post (40" up) to back leg; trim to fit
D1	Deck side angles	2	36"	On top of the arms, along each side of the deck
D2	Deck end angles	2	27"	Under the deck ends, between the D1s
U1	Rail uprights	4	39"	At the 4 corners
R1	Top rails	2	36"	36" above the deck, both sides

Plate and plywood

Stock	Piece	Qty	Size
1/4" plate	Corner plates (base rail to back legs)	2	6 x 6"
1/4" plate	Post cap plates	2	4 x 4"
1/4" plate	Leg cap plates (under C1)	2	4 x 4"
1/4" plate	Gussets (arm to leg), triangles	4	6 x 6"
1/4" plate	Roller yoke plates	4	4 x 8"
2 x 2 x 3/16" angle	Clips	12	3" long
3/4" plywood	Deck	1	36 x 30"

Holes: 3/8 inch bolts go in 25/64 inch holes, the 1/2 inch axles and eye bolt in 33/64 inch holes, and the park pins in 11/16 inch holes. Drill most holes at assembly: clamp the two pieces where they go and drill through both.

5. Parts list

Group	Part	Qty	Rough cost
Lift	Electric hoist, 120 V, 440/880 lb, with pendant, brake, upper limit and double-line pulley block (Harbor Freight type)	1	$110-170
Lift	3 in poly rollers with bearings, for 1/2 in axles	8	$40-60
Lift	1/2 in forged shoulder eye bolt (not a bent-wire one)	1	$10
Lift	5/8 in x 6 in pins with rings + lanyards (park pins)	2	$15
Steel	2 x 2 in square tube 1/8 in wall, 20 ft	2	$190-240
Steel	1-1/2 x 1-1/2 x 3/16 in angle, 20 ft	2	$80-110
Steel	2 x 2 x 3/16 in angle (clips), 3 ft; 1/4 in plate offcuts		$50-70
Bolts	3/8 in Grade 8 bolts, nylock nuts, washers (about 80); 1/2 in x 4 in Grade 8 axle bolts (8)		$60-80
Base	15 in ground auger anchors + chain + turnbuckles	2	$40-60
Deck	3/4 in exterior plywood (half sheet), anti-slip paint, rubber pads		$60-80
Deck	Rubber curb ramp, about 3 in high	1	$25-40
Deck	Ratchet strap; 2 chains with snap hooks across the open ends		$20
Power	Weatherproof box, 20 A switch, outlet, lockable cover	1	$35-50
Power	12 AWG outdoor extension cord, 25 ft	1	$35
Finish	Primer + paint, "NO RIDERS" sign		$40

> **Cost:** about **$800-1,050** in new parts. Scrap-yard steel can save $100-150. No machine shop is needed.

6. Build steps

1. **Plan.** Pick firm, level ground at the end of the back porch. The porch is 30 inches high. Call 811 (free) before screwing in ground anchors. Buy the hoist first, and check its manual for the double-line setup and how its upper limit works.
2. **Cut and paint.** Cut the steel to the lists (or have the supplier do it). Deburr, prime and paint. Don't paint the post faces where the rollers run.
3. **Base.** Bolt the back legs B2 to the base rail B1 with a corner plate on top and a clip inside each joint. Check it's square by measuring the diagonals.
4. **Posts.** Stand the posts P1 on the base rail, 26 inches apart centre to centre, with a clip front and back of each. Bolt the knee braces K1 from each post, 40 inches up, to its back leg. Check the posts are plumb both ways.
5. **Top beam.** Bolt a cap plate on each post top with two clips, then bolt the top beam T1 on the caps. Hang the hoist under the middle of the beam with its own mounting bracket or U-bolts.
6. **Carriage.** Bolt each carriage leg L1 on the back end of an arm A1 with a gusset on each side. Bolt the top tube C1 across the leg tops with the leg cap plates. Drill C1 in the middle and fit the eye bolt, eye up, nut and washer under.
7. **Rollers.** Bolt a yoke plate to the outside of each leg, at the bottom and near the top (Figure 4A). Stand the carriage in place on blocks, then fit the rollers so each one just touches its post.
8. **Deck and rails.** Bolt the D1 angles on the arms and the D2 angles across the ends, then bolt the plywood down and paint it with anti-slip paint. Bolt on the four uprights U1 and the top rails R1. Hang a chain with snap hooks across each open end.
9. **Cable.** Hook the hoist's pulley block on the eye bolt and set up the double line as its manual shows. Slide the hoist's limit stop on the cable so it trips when the deck is level with the porch.
10. **Park pins.** Run it to porch level, mark the posts just under the bottom yoke plates, and drill the 11/16 inch pin holes through both posts. Tie each pin to its post with its lanyard.

11. **Set it up.** Put the lift at the porch with a gap of 1 inch or less between the deck and the porch. Screw in the anchors behind the back legs and tighten the chains. Bolt the right-hand post to the porch frame with a short angle. Put the ramp at the ground end, add the pendant hook on the porch-end post at 48 inches high, and mount the lockout box.

7. Testing

1. **Empty.** Run it up and down a few times. Check the top limit stops it level with the porch, and that the STOP button works.
2. **Load test.** Put 450 lb of sandbags on the platform (1.5 times the 300 lb limit). Run it up and down 10 times, then leave it hanging for 10 minutes: it must not creep. Then set it on the park pins.
3. **Check.** Afterwards, check every bolt and the cable.

8. Using it

1. Turn on SW1. Roll the loaded hand truck up the ramp onto the platform, stand it upright, strap it to the rails and hook the end chains.
2. From the ground, hold UP until it stops at the top. Push in both park pins, then tap DOWN until it sits on them.
3. Go up onto the porch, unhook the porch-end chain and the strap, and roll the hand truck off.
4. To go back down: tap UP to lift it off the pins, pull both pins, then hold DOWN. Let go as soon as it touches the ground. Turn off SW1 and lock it.

9. Upkeep

- **Every month:** look along the whole cable. **Replace it if you see broken wires, kinks or crushed spots.** Check the hook latch, the eye bolt and the anchors.
- **Every 6 months:** check every bolt is tight, and look for rust and cracks. Wipe the post faces.
- **Moving the camper:** lower it, unplug it, take the hoist off and remove the anchors. It unbolts into the base, mast and carriage.